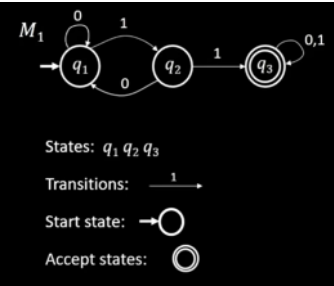


WEEK 0 - LECTURE 0

FINITE AUTOMATON



$r =$	0	1
δ_1	0	0
δ_2	1	1
δ_3	0	1

$\delta: Q \times \Sigma \rightarrow Q$
 $\delta(q, a) = r$ means $q \xrightarrow{a} r$

language

finite string

Example

Regular Language

regular if some finite automaton recognizes

Language

Irregular Language

String

Special String

ϵ is an empty string with a length of zero

Special Language

- $\emptyset / \{\}$ is an empty language set with no strings
- $\emptyset$ has the cardinality/length of zero and not 1, the language with an empty string only has a length of 1

REGULAR EXPRESSION

Regular Operations

Binary Operations

- Union:
- Concatenation:

Unary Operations

- Star Operation:

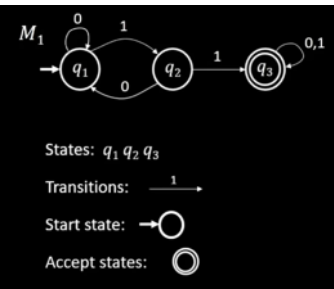
$\emptyset^* = \{\epsilon\}$ empty language star = a language with just the empty string

Closure Properties

- Theorem - Closure under $\cup$
- Theorem - Closure under $\star$

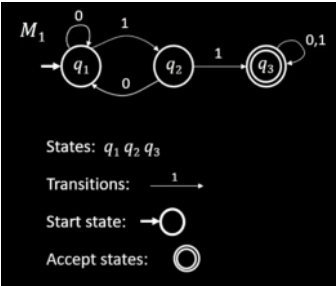
WEEK 0 - LECTURE 1

FINITE AUTOMATON



WEEK 0 - LECTURE 1

FINITE AUTOMATON



finite string

Example

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REGULAR EXPRESSION

Regular Operations

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[Theorem - Closure under U:](#)

[Theorem - Closure under ...](#)